

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2015

Programme: BE

Full Marks: 100

Course: Digital Signal Analysis and Processing

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Write the advantages of digital signals over analog signals. Define Energy and Power signals. 8 1
- b) Show that the impulse response of a BIBO stable LTI system is absolutely summable. 7 2
2. a) Find the convolution between two signals $x[n] = u[n]$ and $h[n] = a^n u[n]$, $0 < a < 1$ 8 2
- b) Find the Z-transform of $x[n] = a^n \cos \omega_0 n u[n]$ 7 3
3. a) Find the 8-point DITFFT of $x(n) = \sin \frac{3\pi n}{4}$, for $0 \leq n \leq 7$ 8 4
- b) What is Zero Padding ? Find the circular Convolution of $x_1[n] = \{1, 2\}$ & $x_2[n] = \{3, 2, 1\}$ 7 4
4. a) Obtain the parallel form realization of following IIR filter 7 5

$$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

- b) Convert the following IIR filter into lattice ladder structure 8 5

$$H(z) = \frac{1 + z^{-1} + 2z^{-2} + z^{-3}}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

5. a) Design a lowpass filter which will have -3 dB cut off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec. 8
- b) Design a linear FIR filter using Kaiser window to meet the following specifications: 7 613
- $0.99 \leq |H(e^{j\omega})| \leq 1.01$, for $0 \leq |\omega| \leq 0.19\pi$
 $\leq |H(e^{j\omega})| \leq 0.01$, for $0.21\pi \leq |\omega| \leq \pi$
6. a) Determine $H(z)$ using impulse invariance method for the following analog 7 7.2

filter's transfer function :

$$H_a(s) = \frac{1}{(s + 0.5) + (s^2 + 0.5s + 2)}$$

7. b) Design a second order discrete-time Butterworth filter with cut-off frequency of 1 KHz and sampling frequency of 10^4 samples/sec by using bilinear transformations. Also plot the poles of the filter.
7. Write short notes on: (Any two)
- 3.2 a) Region of Convergence
 - 2 b) Causal and Non-Causal system
 - 6.1 c) Gibbs Phenomena